Operating Systems
Week 12
Lecture/Discussion 12

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Previous

- Thrashing, What it is and Why it happens

Activity

- Simulating Thrashing, Memory Mayhem

Thrashing

- **4 students as "Processes"** (P1 to P4)
- **1 student as "Memory Manager“**
- **6 spaces on the board labeled as "RAM" (you can draw boxes or use tape)**
- **Sticky notes or small cards with page names (e.g., A, B, C, D)**
- **A list of page requests for each process**

Thrashing activity

each process a list of **4 pages they need**:

- P1 → A, B, C, D
- P2 → E, F, G, H
- P3 → I, J, K, L
- P4 → M, N, O, P

- `[RAM 1] [RAM 2] [RAM 3] [RAM 4] [RAM 5] [RAM 6]`

Thrashing activity

Two Processes (Smooth Execution)

P1 and P2 only

P1 requests Page A

Memory Manager writes "A (P1)" in RAM 1.

P2 requests Page E

Memory Manager writes "E (P2)" in RAM 2.

P1 → B → write "B (P1)" in RAM 3

P2 → F → write "F (P2)" in RAM 4

P1 → C → RAM 5

P2 → G → RAM 6

[A] [B] [C] [E] [F] [G]

Ram is Full, right ?

all 6 RAM slots are full.

Thrashing activity

Add More Processes (Thrashing Starts)

Now **P3 and P4**, and continue the page requests.
P3 requests Page I.

Memory is full, so the Memory Manager must **evict(remove)** a page.
Let's say RAM 1 (A) is removed, and "I (P3)" is added.

P4 requests M.

Evict RAM 2 (E), and write "M (P4)" there.

Note: **Evicting** means **removing a page from memory (RAM)** to make space for another page that is needed.

Thrashing Activity

Now, again when **P1** requests D:

•A, B, C are already **evicted (removed)**, so D will need to replace another one.

•P1 originally needed: **A, B, C, D**

- A is **gone** (evicted by P3)
- B is **gone**
- C is **gone**
- And now **P1 requests D**, which was **never loaded before**

•More page faults begin.

Thrashing Activity

- Since **D is not in RAM**, it's a **page fault**
- because RAM is still full, D must **replace (evict) another page** (e.g., I, J, M, etc.)

- So Whenever Thrashing happens, it is more of Chaos

End result analogy

It’s like 4 people trying to use 6 lockers but each needs 4 items stored. As they keep swapping their stuff in and out, they never find what they need — always going back to the storage room (disk) = **page fault every time**.

Remember!
If People are not laughing at your Goals, that means your goals are too small.